

Jai Moondra

Postdoctoral Fellow at Carnegie Mellon University | Pittsburgh, PA

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Research Profile

Researcher in discrete optimization, responsible AI, online learning, and quantum computing.

Research interests: approximation algorithms, stochastic optimization, algorithmic fairness, machine learning, AI personalization, learning-augmented algorithms, and quantum optimization.

Academic Appointments

Carnegie Mellon University, Tepper School of Business <i>Postdoctoral Fellow. Supervisor: Dr. R. Ravi</i>	Jan 2026 - present <i>Pittsburgh, PA</i>
Massachusetts Institute of Technology, Sloan School of Management <i>Visiting Student</i>	Fall 2024 <i>Cambridge, MA</i>
Georgia Institute of Technology <i>Graduate Research Assistant</i>	Jan 2021 - Dec 2025 <i>Atlanta, GA</i>
Duke University <i>Visiting Research Scholar. Supervisors: Dr. Debmalaya Panigrahi and Dr. Rong Ge</i>	May - July 2018 <i>Durham, NC</i>

Education

Georgia Institute of Technology <i>Ph.D., Algorithms, Combinatorics, and Optimization</i>	2025 <i>Atlanta, GA</i>
• Dissertation: <i>New Directions in Multi-Objective Optimization with Applications</i> • Advisors: Dr. Swati Gupta and Dr. Mohit Singh	
Indian Institute of Technology Delhi <i>B.Tech., Computer Science and Engineering</i>	2019 <i>New Delhi, India</i>

Publications

- Jai Moondra**, Ayela Chughtai, Bhargavi Lanka, and Swati Gupta. “Why Global LLM Leaderboards Are Misleading: Small Portfolios for Heterogeneous Supervised ML.” *Under preparation*, 2026.
- Cheol Woo Kim, **Jai Moondra**, Roozbeh Nahavandi, Andrew Perrault, Milind Tambe, and Swati Gupta. “Many Preferences, Few Policies: Towards Scalable Language Model Personalization.” *Under conference review*, 2026.
- Rohan Ghuge, **Jai Moondra**, and Mohit Singh. “Stochastic Function Certification with Correlations.” *Symposium on Discrete Algorithms (SODA)*, 2027 (to appear).
- Swati Gupta, **Jai Moondra**, and Mohit Singh. “Improved Regret Guarantees for Online Mirror Descent Using a Portfolio of Mirror Maps.” *Under journal review*, 2026.
- Jai Moondra**, Phillip C. Lotshaw, Greg Mohler, and Swati Gupta. “Promise of Graph Sparsification and Decomposition for Noise Reduction in QAOA: Analysis for Trapped-Ion Compilations.” *Quantum*, 2026.
- Swati Gupta, **Jai Moondra**, and Mohit Singh. “Balancing Notions of Equity: Trade-offs Between Fair Portfolio Sizes and Achievable Guarantees.” *Mathematical Programming*, 2026; *Symposium on Discrete Algorithms (SODA)*, 2025.
- Swati Gupta, **Jai Moondra**, and Mohit Singh. “Provably Small Portfolios for Multiobjective Optimization with Application to Subsidized Facility Location.” *Under journal review*, 2025.
- Cheol Woo Kim, **Jai Moondra**, Shresth Verma, Madeleine Pollack, Ling kai Kong, Milind Tambe, and Swati Gupta. “Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning.” *International Conference on Machine Learning (ICML)*, 2025.
- Swati Gupta, **Jai Moondra**, and Mohit Singh. “Which L_p Norm Is the Fairest? Approximations for Fair Facility Location Across All p .” *Economics and Computation (EC)*, 2023.

Reuben Tate, **Jai Moondra**, Bryan Gard, Greg Mohler, and Swati Gupta. “Warm-Started QAOA with Custom Mixers Provably Converges and Computationally Beats Goemans–Williamson’s Max-Cut at Low Circuit Depths.” *Quantum*, 2023.

Jai Moondra, Aditya Sahdev, and Amitabha Tripathi. “Exact and Approximate Results on the Least Size of a Graph with a Given Degree Set.” *Discrete Applied Mathematics*, 2023.

Majid Farhadi, **Jai Moondra**, Prasad Tetali, and Alejandro Toriello. “Multi Purpose Routing: New Perspectives and Approximation Algorithms.” *arXiv preprint*, 2022.

Joel Rajakumar, **Jai Moondra**, Bryan Gard, Swati Gupta, and Creston D. Herold. “Generating Target Graph Couplings for QAOA from Native Quantum Hardware Couplings.” *Physical Review A*, 2022.

Jai Moondra, Hassan Mortagy, and Swati Gupta. “Reusing Combinatorial Structure: Faster Iterative Projections over Submodular Base Polytopes.” *NeurIPS*, 2021.

Internships

Amazon

Research Scientist / Applied Science Intern

Summer 2024

Seattle–Bellevue, WA

- Developed methods to integrate simulation-based and machine-learning predictions for supply-chain optimization.

Duke University, Department of Computer Science

Visiting Research Scholar

Summer 2018

Durham, NC

Teaching

- **Teaching Assistant, Georgia Tech:** supported two graduate algorithms courses; designed homework problems, held weekly office hours, facilitated course discussions, and introduced research-level optimization material to students with varied backgrounds.
- Proposed teaching courses: Design and Analysis of Algorithms, Discrete Mathematics, Combinatorial Optimization, Online Optimization, Stochastic Algorithms, Linear Programming, LLM Optimization, and Quantum Computing.

Selected Talks

- *Balancing Notions of Equity* at SODA, 2025, and IIT Delhi, IISc Bengaluru, and IIT Bombay, 2025.
- *Hybrid Quantum-Classical Algorithms for Max-Cut* at TIFR Mumbai, 2025.
- *Bridging Theory and Practice for Balancing Notions of Equity with Small Portfolios*, INFORMS, 2024.
- *Portfolio Approximations and Fairness in Combinatorial Optimization*, INFORMS, 2023.
- *Which L_p Norm Is the Fairest? Approximations for Fair Facility Location Across All p* , EC, 2023.
- *Fairness Objectives in Facility Location Problems*, INFORMS, 2022.

Awards and Fellowships

- **Honorable Mention, Best Poster**, Mixed Integer Programming Workshop, 2022, for *Reusing Combinatorial Structure: Faster Iterative Projections over Submodular Base Polytopes*.
- **ARC-ACO Fellowship**, Georgia Tech Algorithms and Randomness Center, Spring 2023.
- **National Talent Search Examination (NTSE) Fellowship** for high-school students in India, 2013.

Academic Service

- Reviewer for FOCS, STOC, SODA, NeurIPS, ITCS, IPCO, APPROX, ACDA, *Mathematical Programming, Operations Research*, and SIAM Journal on Discrete Mathematics.
- Session co-chair, Discrete Optimization, INFORMS Annual Meeting, 2023 (with Dr. Swati Gupta).